

649T: 1. (a) $\therefore \log p(x) = ax + b$

$$p(x) = e^{ax+b} = e^b \cdot e^{ax}$$

pdf: $x \in [0, +\infty)$, it must integrate to 1.

$$\int_0^{+\infty} e^b e^{ax} dx = \frac{e^b}{a} e^{ax} \Big|_0^{+\infty} = 0 - \frac{e^b}{a} = 1$$

$$\therefore e^b = -a$$

Thus: $p(x) = -ae^{ax}$, $x \geq 0$

which is the pdf of exponential distribution with rate $\alpha = -a > 0$

(b) Yes, $p(x)$ is a mixture model, because the distribution of x depends on a latent categorical variable z , and we obtain the marginal distribution of x by summing over z .

That's the standard formulation of a mixture model.

$$p(x) = \sum_{k=1}^3 p(z=k) \text{Exp}(x|\alpha_k)$$

$$p(x) = 0.5 \cdot \text{Exp}(x|7.8) + 0.4 \cdot \text{Exp}(x|0.6) + 0.1 \cdot \text{Exp}(x|4.5)$$

$$\text{Exp}(x|\alpha) = \alpha e^{-\alpha x} \text{ for } x \geq 0$$

$$\therefore p(x) = 0.5(7.8e^{-7.8x}) + 0.4(0.6e^{-0.6x}) + 0.1(4.5e^{-4.5x})$$

(c) Yes: $SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$, $\hat{y}_i$ are predicted output.

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$
, $\bar{y}$ are average output.

$$R^2 = 1 - \frac{SSE}{SST} \quad \text{if } SSE > SST,$$

Then $R^2 < 0$

This means, the model predicts the response worse, in squared-error terms, than the constant baseline $\hat{y}_i = \bar{y}$

It can occur, for example, with a poorly fitted model, a model fitted without an intercept, or evaluation on data different from the fitting data.



2. HMM:

$$P(Z[1:2], x[1:2]) = P(Z[1], x[1], Z[2], x[2])$$

$$= P(Z[1]) \cdot P(x[1] | Z[1]) \cdot P(Z[2] | x[1], Z[1]) \cdot P(x[2] | Z[2], x[1], Z[1])$$

$$1) P(Z[1]) = \pi(Z[1])$$

$$2) P(Z[2] | Z[1], x[1]) \rightarrow P(Z[2] | Z[1]), P(x[2] | Z[2])$$

$$\therefore P(Z[1:2], x[1:2]) = \pi(z_1) \cdot P(x_1 | z_1) \cdot P(z_2 | z_1) \cdot P(x_2 | z_2)$$

$$a) (i) x[1] = 0_2, x[2] = 0_1$$

$$\textcircled{1} z_1 = s_1, z_2 = s_1$$

$$P = \pi(s_1) \cdot P(0_2 | s_1) \cdot P(s_1 | s_1) \cdot P(0_1 | s_1) \\ = \frac{1}{2} \cdot (1-p) \cdot a \cdot p$$

$$\textcircled{2} z_1 = s_1, z_2 = s_2$$

$$P = \pi(s_1) \cdot P(0_2 | s_1) \cdot P(s_2 | s_1) \cdot P(0_1 | s_2) \\ = \frac{1}{2} (1-p) \cdot (1-a) \cdot q$$

$$\textcircled{3} z_1 = s_2, z_2 = s_1$$

$$P = \pi(s_2) \cdot P(0_2 | s_2) \cdot P(s_1 | s_2) \cdot P(0_1 | s_1) \\ = \frac{1}{2} \cdot (1-q) \cdot (1-b) \cdot p$$

$$\textcircled{4} z_1 = s_2, z_2 = s_2$$

$$P = \pi(s_2) \cdot P(0_2 | s_2) \cdot P(s_2 | s_2) \cdot P(0_1 | s_2) \\ = \frac{1}{2} (1-q) b q$$

$$(ii) \text{ The likelihood } P(x[1:2] = 0_2, x[2] = 0_1) = \sum_{Z[1:2]} P(Z[1], Z[2], 0_2, 0_1)$$

$$= \frac{1}{2} (1-p) a p + \frac{1}{2} (1-p) (1-a) q + \frac{1}{2} (1-q) (1-b) p + \frac{1}{2} (1-q) b q$$

$$= \frac{1}{2} [(1-p) \{ap + (1-a)q\} + (1-q) \{(1-b)p + bq\}]$$

b) (i) Across four independent sequences, the transition counts are.

$$N_{11} = 1, N_{12} = 1, N_{21} = 0, N_{22} = 2$$

$$L(a, b) \propto a^{N_{11}} \cdot (1-a)^{N_{12}} \cdot (1-b)^{N_{21}} \cdot b^{N_{22}} = a(1-a)b^2$$

Equivalently, each transition probability is estimated by its observed freq.

$$\hat{a}_{MLE} = \frac{N_{11}}{N_{11} + N_{12}} = \frac{1}{1+1} = \frac{1}{2}, \quad \hat{b}_{MLE} = \frac{N_{22}}{N_{21} + N_{22}} = \frac{2}{0+2} = 1$$



(ii) Let $\Delta = p - q \neq 0$ 求 a, b 使 $L(a, b)$ 最大 $P(X \leq 1) P(X \leq 2)$

$$L(a, b) = P(X \leq 1, X \leq 2)$$

$$= \frac{1}{2} [(1-p)(q+a\Delta) + (1-q)(p-b\Delta)]$$

This likelihood is affine in both a, b , so its maximum over $0 \leq a, b < 1$ occurs at boundary.

as for if $p > q$ then $\Delta > 0$ Thus $a=1$ $b=0$

$p < q$ then $\Delta < 0$ Thus $a=0$ $b=1$

3. Device time to failure

Let n_k be the number of observations equal to k , then

$$\sum_{k=1}^{r+1} n_k = n$$

$$(a) \quad L(\theta) = \prod_{i=1}^{r+1} P(X_i | \theta)$$

$$= \left[\prod_{k=1}^r P(X=k | \theta) \right] \times P(X=r+1 | \theta)$$

$$= \prod_{k=1}^r [\theta^{k-1} (1-\theta)]^{n_k} (\theta^r)^{n_{r+1}}$$

$$= \theta^{\sum_{k=1}^r (k-1)n_k + r n_{r+1}} (1-\theta)^{\sum_{k=1}^r n_k}$$

$$l(\theta) = \sum_{k=1}^r (k-1)n_k + r n_{r+1} \log \theta + \left(\sum_{k=1}^r n_k \right) \log(1-\theta)$$

Using $\sum_{k=1}^{r+1} n_k = n$: $l(\theta) = \left(\sum_{k=1}^{r+1} k n_k - n \right) \log \theta + (n - n_{r+1}) \log(1-\theta)$

b) MLE of θ :

Define $A = \sum_{k=1}^{r+1} k n_k - n$ $B = n - n_{r+1}$

$$\frac{dl}{d\theta} = \frac{A}{\theta} - \frac{B}{1-\theta} = 0$$

$$A(1-\theta) = B\theta$$

$$\theta = \frac{A}{A+B}$$

Moreover: $\frac{d^2 l}{d\theta^2} = -\frac{A}{\theta^2} - \frac{B}{(1-\theta)^2} < 0$, $A, B > 0$

so this stationary point is the unique maximum. Finally,

$$A+B = \left(\sum_{k=1}^{r+1} k n_k - n \right) + (n - n_{r+1})$$

$$= \sum_{k=1}^{r+1} k n_k - n_{r+1}$$

$$\hat{\theta}_{MLE} = \frac{\sum_{k=1}^{r+1} k n_k - n}{\sum_{k=1}^{r+1} k n_k - n_{r+1}}$$

